Shadowplay

A lamp on your desk, a story on your wall. Your hands cast the characters.

Entertainment track | New Experience | Target: Best New Entertainment Experience

AT A GLANCE

Track / division	Entertainment (immersive storytelling) / New Experience
Primary prize	Best New Entertainment Experience ($100,000; runner-up $50,000)
Fallback special award	Best First Five Minutes ($25,000)
Session length	One chapter is 5 to 8 minutes; the full story is about 35 minutes over 5 chapters
Room requirement	A wall in front of the seated player, or a floating screen fallback
Core tech	WebXR hand tracking, plane detection (wall), real-time shadow mapping, silhouette matching
Biggest risk	Recognising hand shapes reliably, especially when both hands overlap
Why it wins attention	Everyone already knows how to make a shadow rabbit: zero learning curve, instant delight
Availability assumed	Full time: roughly 300 to 400 hours through 17 Nov

The one-line pitch

An interactive shadow-puppet story performed on your real wall. A virtual lamp on your desk throws the shadows of your real hands across the room; form the right shape and your shadow comes alive as a character, then carries the story forward.

SECTION 01

The idea

Shadowplay is a short interactive story told entirely in shadows on the wall of your own room. You sit at your desk. A small warm lamp appears on the desk in front of you and the room dims slightly, like a theatre before a show. When you raise your hands between the lamp and the wall, their shadows appear on your real wall, stretched and soft, exactly as they would with a real lamp.

The story asks you for characters. A narrator says the field is too wide for a moth to cross alone, and a faint outline of a rabbit appears on the wall. You make the rabbit shape with your hand. When your shadow matches, it comes to life: the ears twitch, it blinks, and it hops across the wall when you move your hand. From then on you are both the audience and the puppeteer.

Why it fits the Entertainment track

The track asks for lean-back media and immersive storytelling with a hands-first implementation. Shadowplay is lean-back in pace (narrated, gentle, no fail states) but the story literally cannot move without the player's hands. It is also one of the oldest forms of storytelling there is, reinvented with technology that can see your fingers.

SECTION 02

The core insight

Shadow puppetry is a hand language almost everyone already speaks. Most people learned a rabbit, a dog or a bird as children. That solves the hardest problem in controller-free design, which is teaching gestures: the player arrives already knowing the vocabulary.

The second insight is that light obeys physics people understand. Move your hand toward the lamp and the shadow grows; move it toward the wall and it shrinks and sharpens. Shadowplay turns that into a puzzle mechanic, so the player's real-world intuition drives the story.

Physical rule	How it becomes gameplay
Closer to the lamp means a bigger shadow	Grow your wolf big enough to scare off the storm, or shrink a mouse small enough to slip through a keyhole.
Shadows overlap and merge	Two hands together form one larger creature, such as a bird with two wings.
Moving the lamp moves every shadow	Pinch and drag the lamp to sweep a searchlight across the wall and reveal hidden things.
Soft versus sharp edges	A sharp, close shadow is 'solid' and can push objects; a soft, far one is a ghost that can pass through them.

SECTION 03

The story: The Moth Who Borrowed Light

An original story written for this project (placeholder title). A small moth has lost its glow to the Long Night and must cross the world on your wall to borrow light from the lighthouse at the edge of the sea. Each chapter introduces a new shape and a new use of the light.

Chapter	Setting and beat	Shapes and mechanics introduced	Length
Prologue	The lamp flickers on; the moth lands on its shade and tells its problem.	Open hand (wave hello); the moth follows your fingertip.	3 min
1. The Field	Tall grass the moth cannot cross.	Rabbit: hop across; move your hand to steer.	6 min
2. The River	A dark river under the moon.	Fish or boat; distance to the lamp controls size, so the boat must be big enough to carry the moth.	7 min
3. The Wolf	A shadow wolf blocks the path.	Dog or wolf; choose: grow huge and scare it, or become a small bird and fly past. A branching choice with two endings for this chapter.	7 min
4. The Storm	Wind and rain on the wall.	Drag the lamp to find shelter; two-hand bird to fly.	6 min
5. The Lighthouse	The keeper is asleep; the light is out.	Cupped hands hold the light; open them to release it across the whole wall. Finale.	6 min

Each chapter ends on a complete moment, so the one bus stop test holds: a player can do one chapter in a short break and feel finished.

SECTION 04

The player experience

The first five minutes

Time	What happens
0:00	Passthrough fades in. A small lamp appears on the desk; the room dims gently except for a pool of warm light on the wall ahead.
0:15	The player's hands naturally enter the light. Their shadows appear on the wall. Most people start making shapes immediately, unprompted.
0:40	A moth flutters onto the lampshade. A narrator (voice plus English captions) says hello. The moth flies to the player's shadow fingertip and follows it.
1:30	The first request: a faint rabbit outline appears on the wall. The player makes the rabbit shape; the outline fills in and the shadow comes alive.
2:30	The rabbit hops wherever the hand moves. The player carries the moth across the field.

Time	What happens
4:30	Chapter 1 ends. The rabbit is added to the Shadow Journal, a small book on the desk that collects every creature discovered.

Interaction vocabulary

Interaction	How the player does it	Effect
Cast a shape	Form a hand shape in the lamp light and hold it for half a second.	Matching shape comes alive as a character.
Puppeteer	Move, tilt or open and close the hand.	The character moves; opening and closing fingers moves its mouth or wings.
Resize	Move the hand toward or away from the lamp.	The character grows or shrinks.
Move the lamp	Pinch the lamp and drag it along the desk.	Every shadow and the light pool shift across the wall.
Release	Open the hand fully and pull it out of the light.	The character bows and fades; the hand is free again.
Pause	Turn the lamp off with a pinch on its switch, or take the headset off.	The story pauses; captions show where you left off.

SECTION 05

Why passthrough and the real room matter

- **Your wall is the stage.** The story is projected onto the real wall in front of you, at the real size of that wall, lit by a lamp on your real desk. With passthrough off there is no wall and no theatre.
- **Your real hands cast the shadows.** The characters are made from your own hands, which you can see in passthrough at the same moment as their shadows.
- **The room dims like a theatre.** A WebXR app cannot dim passthrough directly, so the app draws a large translucent dark shell around the player with a soft-edged opening where the stage is. Passthrough shows through it darker everywhere except the light pool: a moment of ceremony that turns an ordinary room into a stage.
- **The stage adapts to the wall.** The light pool and the story's scene fit the detected wall size and distance. A close wall gives an intimate show; a far one gives a big-screen show.

Holding up in rooms the developer never tested

The app looks for the wall plane most directly in front of the seated player, between about 0.6 and 4 metres away. **The most common case is a desk pushed against a wall**, so the wall is often only 0.6 to 1 metre away and partly blocked by a monitor. For close walls, the stage sits higher on the wall, above the monitor, and is scaled down; the lamp moves closer to the player so shadows stay large enough. If no suitable wall is found (the player faces an open room or a window), it places a softly glowing paper screen floating 2 metres ahead. The story plays identically in all three cases, so no room blocks the experience.

SECTION 06

Seated, FoV-aware design

- The stage is about 1.2 m wide at a distance of about 2 m, roughly 33 degrees across, which fits inside the Quest 3S field of view without head turning. On a close wall behind a monitor (about 0.8 m), it shrinks to about 0.5 m wide to keep the same angle.
- Hands work in a comfortable zone just above the desk, between the lamp and the wall. Elbows can rest on the desk.
- Captions sit at the bottom of the stage, never at the edges of the view.
- No standing or walking. The whole show happens from a chair.

SECTION 07

Shape recognition: the hard part

Shadowplay uses two complementary methods to decide whether the player's hand matches a target shape.

Method 1: joint-angle templates

WebXR hand tracking reports 25 joints per hand. For each frame the app computes a compact feature vector: how curled each finger is, how far apart the fingers are spread, where the thumb tip is relative to the other fingers, and which way the palm faces relative to the lamp. It compares this vector with stored templates for each shape and accepts a match within a tolerance. This is cheap enough to run every frame.

Method 2: silhouette matching

Because what matters is the shadow, not the exact hand pose, the app can also compare shadows directly. A few times per second it renders the hands' shadow into a tiny 64 x 64 black-and-white image and measures how much it overlaps the target silhouette. A match is accepted when the overlap is above a threshold (starting around 55%) for half a second. This is forgiving in exactly the right way: any hand pose that looks like a rabbit counts as a rabbit, even if the fingers are arranged differently.

> **Design rule: choose shapes that hand tracking can see**
>
> Hand tracking degrades when fingers are hidden behind other fingers or when two hands overlap or interlock. The story is therefore built mainly on **one-hand shapes** (rabbit, dog, duck, snail, deer, fish) and on two-hand shapes where the hands sit **side by side** rather than interlocked (the bird uses two hands next to each other as wings). Every two-hand shape also has a one-hand alternative for accessibility.

SECTION 08

Progression and reasons to come back

System	Description
Chapters	Five chapters, each a complete moment. Players return to see what happens next.
The Shadow Journal	A small book on the desk that collects every creature discovered, with a page showing its shape. Hidden creatures (swan, snail, crocodile, deer) can be found by experimenting.
Shape of the Day	A daily riddle ('I carry my house on my back') with a new shape to discover. Solving it adds a page to the journal.
Free Stage	After the story, an open stage mode where any discovered creature can be performed freely with music, for play or to show friends.
Encore endings	Chapter 3's branching choice gives a reason to replay.

SECTION 09

Accessibility

- Adjustable recognition tolerance, from strict to very forgiving.
- A one-hand alternative for every two-hand shape.
- Full English captions for all narration, placed on the stage.
- No time pressure and no fail states; the story waits for the player.
- A 'show me' option that briefly displays a ghost hand demonstrating the shape, so no one gets stuck.
- Optional voice alternative: if speech recognition is available in Quest Browser, saying the creature's name also summons it (needs verification on device).

SECTION 10

Art direction and sound

Look

Warm lamplight on a real wall, deep soft shadows, and characters drawn as elegant silhouettes with small animated details (a blinking eye, fluttering ears). The palette is amber, ink black and moonlight blue. Silhouette art is forgiving and quick to make, which suits a solo schedule: every character is a 2D shape with a few animated parts.

Sound

- A music-box and soft strings score, **commissioned or properly licensed**. Self-composed music is only competitive if you already produce music.
- Narration by a **commissioned voice actor**, credited in the submission. Text-to-speech only with a licence that allows commercial, competition and promotional use.
- **Licensing for a web app is stricter.** The audio files are publicly downloadable from the hosted site, so licences must allow distributing them, not just playing them. Use work-for-hire agreements for commissions and keep every licence and release in the repo.
- Send the voice and music briefs in week 1: commissioned work takes weeks to turn around.
- Spatial audio: sounds come from the wall where the character is, and the lamp hums faintly from the desk.
- Each creature has a small signature sound when it comes alive.

SECTION 11

Technical architecture (IWSDK)

Feature	Used for
Hand tracking	Joint poses for shape recognition; a simple hand mesh built from joints casts the shadows.
Plane detection	Finding the wall in front of the player (semantic label 'wall').
Shadow mapping	One spot light (the lamp) with a 1024 x 1024 shadow map; the wall is a transparent surface that only shows shadows.
Render-to-texture	The 64 x 64 silhouette buffer for matching, updated a few times per second rather than every frame.
Anchors	Keeping the lamp and stage fixed in the room.
Web Audio / spatial audio	Narration, score and positioned sound effects.

System	Responsibility
StageSetupSystem	Chooses the wall or the floating screen fallback; places the lamp on the desk.
HandCasterSystem	Updates the hand shadow meshes from joint poses every frame.
ShapeMatchSystem	Runs joint-angle matching each frame and silhouette matching at about 5 Hz; emits 'shape matched' events.
PuppetSystem	Once matched, maps hand motion to character motion and animation.
StorySystem	A state machine driven by a JSON script of beats, requests and branches.
NarrationSystem	Plays narration and shows captions in sync.
JournalSystem	Records discovered creatures in IndexedDB.

Performance notes

- One shadow-casting light only; keep the shadow map at 1024 pixels.
- Reading pixels back from the GPU is slow and can stall a frame, so the silhouette check runs on a tiny buffer at a low rate using asynchronous readback (a WebGL2 pixel buffer plus a fence) where available, and joint-angle matching carries most of the work.
- WebXR reports hand joints but no hand mesh. The shadow caster is either simple capsules along the joints or a skinned hand model bundled with the app (never loaded from an external CDN at runtime). The week-1 spike decides which looks convincing.
- Characters are flat textured quads with a few bones, so drawing cost is tiny.

SECTION 12

Risks and mitigations

Risk	Severity	Mitigation
Shape recognition is unreliable	High	Two methods combined; generous tolerances; shapes chosen for trackability; 'show me' helper. Tested in week 1.
Two-hand shapes break tracking	High	Avoid interlocking shapes; side-by-side only; one-hand alternatives.
The story is not compelling	Medium	Write the full script in week 1; test it as a read-aloud with friends before building.
Narration and music quality	Medium	Commission a voice actor and composer with distribution rights; keep the score simple.
No wall in front of the player	Low	Floating paper screen fallback.
Shadow rendering cost	Low	Single light; small shadow map; tested early on device.
The wall is right behind the monitor	High	The most common desk setup. Stage placed above the monitor and scaled; lamp moved closer; tested in the playtest rooms.
Joint-capsule shadows look like sausages, not hands	Medium	Week-1 spike compares capsules with a bundled skinned hand model; pick whichever reads as a real hand shadow.
Silhouette readback stalls frames	Medium	Tiny buffer, low rate, asynchronous readback; joint templates as the primary method.
Passthrough cannot be captured for the video	High	Verify on day 1 that Quest recording or casting captures passthrough with the WebXR layer; if not, plan an alternative before week 4.
Tracking quality varies with lighting and skin tone	Medium	Playtest across varied lighting, hand sizes and skin tones; tune recognition on the whole set, not the developer's hands.
Commissioned audio arrives late	Medium	Brief the voice actor and composer in week 1; build with placeholder audio; final audio due by week 5.

Week 1: spikes

More hours reduce scope and content risk, but not platform risk: hand-tracking accuracy and how convincing a rendered shadow looks do not improve with effort. Week 1 measures those before the story is built.

Day	Spike	What to measure
1	Passthrough capture check (first hour)	Record a short clip of a WebXR scene over passthrough using Quest recording or casting. If it fails, the video plan changes.
1 - 2	Shadow render	Cast hand shadows from a lamp onto the detected wall, comparing joint capsules with a bundled skinned hand model. Judge on device whether they read as real hand shadows. Test a close wall behind a monitor and a far wall.
2 - 4	Shape recognition	Record templates for 8 one-hand shapes and 2 side-by-side two-hand shapes. Measure recognition accuracy across at least 5 people with different hand sizes and skin tones, for joint templates alone and with silhouette matching added.
5	Story and audio	Full script table read with two or three listeners; send the voice-actor and composer briefs.

Go criteria for Shadowplay

Shadows read as real hand shadows on device, on both a close and a far wall. At least 90% recognition for the one-hand shapes across all testers, with no false matches while hands are simply moving through the light. Listeners want to know what happens after the prologue.

Decision order

1. **All pass:** build Shadowplay at full scope.
2. **Recognition below target for some shapes:** drop or replace those shapes and loosen tolerances (about 1 day). The story is written around shapes, so swapping one is cheap in week 1.
3. **Two-hand shapes unreliable:** make every two-hand moment optional, with its one-hand alternative as the default.
4. **Shadows do not look convincing with either method:** this is the concept's core, so reconsider the idea choice before building further.

SECTION 13

Seven-week build plan

Week	Dates	Goal	Exit criterion
Week 1	29 Sep - 4 Oct	Spikes: capture, shadow render, recognition across 5+ people; script table read; commission voice and music	Go criteria met; capture method confirmed; briefs sent
Week 2	5 - 11 Oct	Stage setup (close wall, far wall, floating fallback); room dimming; lamp grab; recognition system; puppeteering	A matched rabbit hops around the wall under hand control in all three stage cases
Week 3	12 - 18 Oct	Story engine (JSON beats); Prologue and Chapter 1 with placeholder audio; first five minutes	First 10 minutes playable end to end
Week 4	19 - 25 Oct	Chapters 2 and 3 including the branch; Shadow Journal; **start capturing video footage**	Half the story complete; first usable footage
Week 5	26 Oct - 1 Nov	Chapters 4 and 5; final narration and score integrated; Shape of the Day; Free Stage	Content-complete for the must-ship list
Week 6	2 - 8 Nov	Playtests in 10+ rooms (lighting, skin tones, hand sizes, desk-against-wall, no wall); accessibility options; performance pass	72 fps held; no tester gets stuck; every room gets a stage
Week 7	9 - 15 Nov	First-five-minutes iteration until every cold tester succeeds; animation and sound polish; final video edit; submission text	Video cut from weeks of footage; build frozen candidate
Ship	16 - 18 Nov	Final checks on a clean device; submit on 17 Nov	Submitted a day early; hosted link verified

Where the extra full-time hours go

The content load (script, voice direction, five chapters) now fits the schedule. Beyond that, spend hours where judges score: **playtests** in 10 or more real rooms, **iterating the first five minutes** until every cold tester makes their first shape unprompted, and **video footage** captured from week 4 onward.

SECTION 14

Scope

Must ship	Nice to have	Cut
Wall detection with close-wall handling and floating fallback; room dimming; lamp; 8 to 10 shapes; all five chapters including the branching choice; commissioned narration and score; captions; Shadow Journal; Shape of the Day; Free Stage; tolerance, one-hand and 'show me' options	Voice alternative for summoning creatures (if speech recognition works in Quest Browser); more hidden creatures	Multiplayer shadow theatre, user-authored stories and accounts: a backend adds failure modes during the judging window and adds nothing to the score. In-app video recording (the Quest's own capture covers it).

SECTION 15

Demo video plan (under 3 minutes)

Time	Beat	Shot
0:00 - 0:12	Hook	A dim room, a lamp clicks on, a hand rises and a shadow rabbit on the real wall blinks and hops away.
0:12 - 0:40	Setup	Wide shot showing the real wall, desk and lamp; the room dimming like a theatre.
0:40 - 1:30	Play	Real-time Chapter 1: the moth, the rabbit request, the shape match, puppeteering.
1:30 - 2:10	Physics as puzzle	Moving the hand to the lamp to grow the wolf; dragging the lamp to reveal a hidden path.
2:10 - 2:40	Discovery	Flipping through the Shadow Journal; a hidden snail found by experimenting.
2:40 - 2:55	Close	Finale light spreading across the wall; title.

SECTION 16

Submission draft

Name	Shadowplay
Tagline (under 140 characters)	An interactive shadow-puppet story on your real wall. Your hands cast the characters, and the story waits for them.
Track / division	Entertainment / New Experience
Target launch	Web release Q1 2027 with additional chapters; Horizon Store PWA after that

Description (first draft)

Almost everyone learned to make a shadow rabbit as a child. Shadowplay turns that shared memory into an interactive story told on the wall of your own room.

A small lamp appears on your real desk and the room dims like a theatre. Your hands cast soft shadows on your real wall. When the story needs a character, you make its shape, and your shadow comes alive: a rabbit that hops where your hand goes, a wolf that grows as you move closer to the lamp, a bird made of two hands. You help a small moth cross the world to borrow light from a distant lighthouse.

We built Shadowplay with the Immersive Web SDK. Plane detection finds the wall in front of you, a real-time shadow map casts your tracked hands onto it, and two recognition methods, joint-angle templates and silhouette matching, decide when your shadow matches a character. There are no controllers, no menus to learn and no text walls: the first shape you make is the tutorial.

SECTION 17

How it maps to the judging criteria

Criterion (25% each)	Evidence in Shadowplay
Innovation and Creativity	An ancient art form reinvented with hand tracking; light physics as a puzzle mechanic.
Experience Design	Pre-learned gestures, no text tutorials, a room that becomes a theatre, chapters sized for short sessions.
Technical Implementation	Wall detection with fallback, real-time shadows from tracked hands, dual recognition methods.
Polish and Presentation	Silhouette art is striking and achievable solo; a hook that reads instantly on video.

Hard requirement and design guidelines

Test	What the rules ask	How this idea passes
Hands-first (hard requirement)	Complete the entire experience without ever pairing a controller.	Every interaction is a hand shape, a hand movement or a pinch on the lamp.
Airplane seat test	Every interaction works within a two-foot radius, seated.	Hands work in the space just above the desk; the stage is directly ahead.
One bus stop test	Fast cold start, clean pause/resume, a satisfying moment in 10 minutes or less.	Each chapter is 5 to 8 minutes and complete; pausing is a lamp switch or removing the headset.
Take it away test	Remove any third-party service and a project still stands.	No third-party services; all content is original and runs locally.

Competition rules that shape this plan

- **One prize per entry.** Rule 7 limits each entry to a single prize, so this brief aims at one track prize and names one special award as a fallback rather than trying to stack awards.
- **Submission:** a public HTTPS link (GitHub Pages, Vercel or similar) that stays live until the winner announcement (about 11 Dec 2026), a public YouTube/Vimeo video under 3 minutes, and the Devpost form (name, 140-character tagline, track, division, description of about 500 words, hand-interaction notes, target launch date).
- **No changes after the deadline.** Judging uses the build as it stood at 18 Nov 2026, 12:00 PT. Plan to submit on 17 Nov and freeze the hosted build.
- **Content rules:** English (or English subtitles), no identifiable real people other than entrants, no brand names or logos, all music, art and fonts properly licensed or original.
- **Video rules:** real footage from a Quest or the XR simulator; do not lean on AI-generated video to carry the pitch.
- **Performance floor:** judges expect at least 60 fps on Quest hardware. Target 72 fps.
- **Voice and music:** keep licences or signed releases for narration and score.

SECTION 18

Honest weaknesses

- Recognition quality decides everything. If shapes are frustrating to trigger, the magic disappears.
- Storytelling quality is harder to guarantee than mechanics; a weak script makes it feel like a tech demo.
- Replay value after the story ends is limited; the journal, daily shape and Free Stage help, but it is a story first.
- Narration and music take real money and lead time; commission them in week 1.
- The desk-against-wall case makes the stage smaller and closer than the ideal; it must still feel like a theatre.

SECTION 19

Scorecard and verdict

Dimension	Score / 10	Comment
Originality	9	Distinctive and instantly understood.
Only possible in MR	8	Needs the real wall and real hands; the fallback screen weakens it slightly.
Works in untested rooms	8	One wall, or the fallback.
Solo feasibility in 7 weeks	8	Full-time hours cover the story content; recognition is the remaining hard part.
Prize ceiling	$100k	Top Entertainment prize; First Five Minutes as a fallback.

Verdict

The best backup to Tabletop Beat, and a strong first choice on its own if you prefer storytelling to games. With full-time hours its content load (script, voice, five chapters) fits the schedule, so the full story can ship. It has the most emotional pull of the six ideas and the strongest first five minutes.